Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building **Python/FastAPI services**, **ETL pipelines**, and **AI-powered full-stack products** across **SQL, Docker, Kubernetes**, and LLM workflows for data-centric production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema, query, and indexing optimization across backend data services.
- Built **FastAPI** integration and job-orchestration workflows across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation checks and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with modular backend services, secure ingestion, and AI-driven grading workflows.
- Designed concurrent services on **AWS RDS** with REST APIs and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

USC AD Labs **Los Angeles, California**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines for autonomous-driving planning, processing **100K+ trajectory logs** and improving long-horizon planning quality by **18%** across data-heavy evaluation workloads.
- Architected an HD-map-aware spatial-temporal encoder in **PyTorch**, improving closed-loop stability by **15%** while strengthening experience with AI systems, model iteration, and performance tradeoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and release flows across **3 operating systems** with **20+ unit/integration suites**.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and Dockerized microservices, supporting **8+ workflows**, automated business logic, and **2M+ synchronized records**.
- Automated **Jenkins** testing and deployments for Azure SQL-backed services, improving release reliability while reducing resource wastage by **42%** and increasing lead conversion by **15%**.

Skills

Languages: Python, JavaScript/TypeScript, Java, C++, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Async Services, Microservices
Frontend: React.js, Vue.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, SQL Server, Azure SQL, Snowflake, SQLAlchemy
Cloud & Infrastructure: Docker, Helm, Kubernetes, Terraform, AWS (RDS, S3), Microsoft Azure, Jenkins, GitHub Actions
Data & AI: ETL Pipelines, Argo Workflows, LLM Orchestration, Agentic AI Systems, RAG, PyTorch, Transformers
Engineering Practices: Distributed Systems, Event-driven Services, Unit & Integration Testing, Technical Documentation, Open-source Collaboration, Agile

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built a cloud-based document platform over **50K+ documents** with RAG-style retrieval, semantic search, and collaborative workflows that reduced review time by **60%**.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, Dockerized deployment, and structured data workflows from a unified interface.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a cloud-native fintech platform supporting **50K+ users** and **1M+ transactions** with containerized services, real-time synchronization, and production-grade deployment workflows.